



Department of Computer Science & Information Technology



Course Code: **CS-11303**

Credit Hours: 4 (3,1)

Batch: **Fall-2025**

Course Title: **Operating System**

Lecture Hours/Week: **3**

Pre-Requisites: **None**

Complete Analysis of Effectiveness of Course and Level of Skills Ensured in Operating System

The Operating Systems Lab ensures students translate theory into practice through Linux-based implementation.

Practical Skill Development

Students develop:

- Linux command-line proficiency
- Process and memory management understanding
- Shell scripting automation skills
- Secure permission management ability
- Basic system programming expertise

Industry-Level Readiness

The lab aligns with real-world roles:

- Linux System Administrator
- Cloud Support Engineer
- DevOps Associate
- SOC Analyst (Linux environment)

Modeling & System Design Competency

Students demonstrate ability to:

- Analyze system behavior
- Optimize resource allocation
- Implement controlled process execution
- Manage user-level security

The Operating Systems course (Theory + Lab) effectively bridges academic learning and industrial system-level expertise. It builds a strong foundation in computing infrastructure, enabling students to pursue advanced study and professional careers in system administration, cloud computing, and cybersecurity domains.